Fiscal Year:	FY 2015	**Task Last Updated:**	FY 10/14/2015
PI Name:	Bloomfield, Susan A. Ph.D.		
Project Title:	Iron Overload and Oxidative Damage: Regulators of Bone Homeostasis in the Space Environment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	77843-4375	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	08/01/2013	**End Date:**	07/31/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2017 per NSSC information (Ed., 5/5/16)		
Key Personnel Changes/Previous PI:	No changes		
COI Name (Institution):	Ford, John (Texas Engineering Experiment Station) Hogan, Harry (Texas Engineering Experiment Station) Turner, Nancy (Texas AgriLife Research)		
Grant/Contract No.:	NNX13AL25G		
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Task Description:	Recent work in bone biology field has demonstrated that cellular damage due to molecules called reactive oxygen species (so-called oxidative damage) is a key mechanism contributing to aging-related bone loss, as well as to that occurring with the loss of estrogen after menopause. We know very little about whether the loss of gravity as experienced in the space environment modulates oxidative damage to bone. A newly recognized source of oxidative damage in bone is that due to iron overload, or an increase in body iron stores, as has now been demonstrated in both International Space Station (ISS) crew members. Data published this month indicate that sub-pathological levels of serum ferritin (a marker for iron stores) contribute to accelerated bone loss and to incident vertebral fractures in healthy middle-aged men and women. We have preliminary data indicating that similar moderate elevations in body iron stores, as induced by a high iron diet, can result in loss of cancellous bone and diminished bone formation activity in rodents. The experiments in this proposal will utilize iron overload as an experimental intervention to exacerbate oxidative damage during a period of simulated microgravity (by hindlimb unloading in male and female mice) to define the sex-specific mechanistic role of oxidative damage as a key regulator of bone mass and, ultimately, fracture risk as modulated by the gravitational unloading of the space environment. In a final specific aim, we will test whether superimposing acute and continuous (over 42 days) radiation exposure exacerbates oxidative damage and bone decrements during hindlimb unloading. These experiments are deliberately designed to enable a rapid transition to flight experiments requiring minimal intervention during flight aboard ISS, should these ground-based simulations yield provocative findings to justify flight experiments. Iron overload can be maintained by dietary intake, and limiting the acute radiation exposure in Experiment 2 to just before the unloading period allows for testing of the same hypotheses in a flight experiment. The experiments outlined in this proposal address how microgravity affects key regulatory mechanisms governing alterations in bone (from OCB-2 in the Space Biology Plan) and also one of the National Research Council (NRC) Decadal Survey's highest priority recommendations, to expand our understanding of sex differences in adaptation to the space environment, particularly in the area of bone and long-term radiation risks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have evidence that oxidative damage caused by molecules that oxidize DNA or other tissue components can contribute to the loss of bone mass and integrity that occurs with aging and with the loss of estrogen in post-menopausal women. There are some unfortunate individuals who experience pathological increases in body iron stores (with diseases like hemochromatosis) who, we believe, also experience increased oxidative damage to multiple tissues due to an excess of free iron molecules. However, we now know that even moderate (sub-clinical) elevations in body iron stores in healthy middle-aged individuals can result in accelerated bone density loss. Only women in these groups, however, exhibit an increased risk (5-fold) of incident vertebral fractures. These experiments in skeletally mature mice will yield important and detailed information on the impact of these moderate elevations in body iron on evidence for oxidative damage and for accelerated bone loss, with the unique added treatment of reduced weight bearing (simulating prolonged bed rest in humans). Because this project will be examining both male and female mice, we will be able to define if there is any sex difference in the response of bone to iron overload, with and without coincident disuse. Our final specific aim will test the impact of increased iron stores in combination with disuse and exposure to chronic, very low dose gamma radiation, simulating space radiation. Radiotherapy patients on Earth experience a 2.6-fold increase in fracture risk compared to age-matched peers. A better understanding of the additional impact of radiation on bone integrity, with and without iron overload, may lead to better protection of bone health in radiotherapy patients, reducing fracture risk.
Task Progress:	Tasks accomplished --Year 2: August 2014: Submitting amendment to animal use protocol summarizing changes in protocol decided upon at end of Year 1 (deleting gonadectomies and hormone replacement by pellet implant, shortening HU--hindlimb unloading--period to 4 weeks) September – December 2014: Completing 2 cohorts of male mice (n ~44) on protocol above; collecting in vivo Piximus scans, storing tissues collected at termination for later analyses. Atomic absorption spectrometry assays were run for liver and serum total iron on samples from Year 1 cohorts to confirm elevated iron stores on our high iron mouse diet. Femurs sent to Dr. Matt Allen, Indiana University, for micro-CT scans. January 14, 2015: Investigator meeting at NASA Human Research Program (HRP) meeting, Galveston, TX (including consultant Sara Zwart, Johnson Space Center: review of interim data, discussion of priority assays moving forward. Preliminary data reported at HRP meeting on Year 1 cohorts (Metzger et al., 2015). January – April 2015: Completed 2 cohorts of female mice (n~ 44) on revised protocol; collecting in vivo Piximus scans, storing tissues collected at termination for later analyses. Femurs sent to Dr. Matt Allen, Indiana University, for micro-CT scans. MicroCT data on male cohorts delivered by Dr. Allen to Principal Investigator (PI); bones returned to College Station for embedding and further analyses. Dr. Turner's group validated all mouse assay kits to be used on serum and liver samples (oxidative damage markers, iron status variables). May 11, 2015: Investigator meeting in College Station, with consultants S Zwart and MR Allen on the phone. Review of interim Piximus and some microCT data, discussion of experiments moving forward, especially need for an interim dose-response experiment. Early planning for continuous radiation exposure experiments. May- July 2015: Ordering mice and 3 high iron mouse diets for dose-response experiment; lack of availability of 4-mo-old mice requires we "age up" mice here for at least 3 weeks before beginning experiment (early August 2015). Mice for pilot studies adding exercise groups to radiation plus HU experiments ordered. MicroCT data on female cohorts delivered by consultant MR Allen. Processing bone samples for bone histomorphometry and for immunohisto-chemical (IHC) staining for oxidative damage marker in bone. Turner lab is batching liver and serum samples, beginning assays for: serum ferritin and hepicidin (iron status); liver SOD, catalase, glutathione peroxidase (oxidative stress markers). Interim Results: As expected, we observed reduced body weights (~11% lower vs. cage control rats) after 4 weeks of hindlimb unloading (HU) even though food consumption increased by about 60% during week 3 of HU. This suggests a reduced food efficiency (nutritional absorption of food in the gut) during HU, which has been observed previously. Validating iron status in our experimental groups: Atomic absorption spectroscopy assays confirmed a 28% elevation in liver iron in mice from Year 1 cohort eating the high iron diet. Unfortunately, results for all but a few serum samples

	tested for total iron content by the consulting laboratory were rendered invalid by technical error. After discussions with the lab's PI, we are cautiously moving forward with liver total iron by a.a. spectrometry assays on this year's cohorts, with assurances that the same technical error will not be repeated. These assays are in progress, as well as ELISAs to assess serum ferritin and serum markers of oxidative stress. Bone outcomes (preliminary): We have confirmed our early results from Year 1, indicating a positive effect of increased iron stores on uCT bone outcomes; this negates our working hypothesis, that elevated iron stores would negatively impact on cancellous bone. All groups consuming the high iron diet had higher distal femur cancellous bone volume (%BV/TV) than matched groups of the same weight bearing status consuming a normal iron diet. Mice with elevated iron stores do lose %BV/TV following 4 weeks of hindlimb unloading (HU), but these mice finish the experiment with cancellous bone mass similar to weightbearing mice on a normal iron diet. Other uCT outcomes describing cancellous bone microarchitecture are consistent with the pattern observed below: HU induces declines in trabecular number and thickness in both diet groups, but mice on the high iron diet have higher values than those on normal iron diet. We have preliminary evidence about the impact of elevated iron stores on a cortical bone site (midshaft femur) in female mice. There is no apparent impact of the high iron diet on mid-shaft bone area, marrow area or cortical thickness in weight bearing cage control mice; however, the reduction in bone area and cortical thickness are blunted in those animals on the high iron diet. Preliminary data from mechanical testing on mid-shaft femurs revealed that ultimate load (a key measure of bone strength) was reduced by 15 and 24% in HU males and females, respectively, is reduced by HU, but elevated iron stores result in stronger bone at this site within weightbearing groups. These differences were statistically significant in males only. [Stiffness and derived material properties will be calculated from the raw data before we present data at the American Society of Bone & Mineral Research in October, 2015.] Ultimate load at the femoral neck, a clinically relevant site, was reduced by 15 and 24% in HU males and females, respectively, while the high iron diet produced a gain, but only in cage control animals. Again, these differences reached statistical significance only in males. Inability to detect significant changes in the female mice mid-shaft ultimate load may be due to greater variability in ultimate load values for the female cage control-high iron (CC+Fe) group. Interestingly, these results indicate few sex differences thus far and, contrary to our original hypothesis, a positive impact of elevated iron stores on bone outcomes in mature adult mice. These results will be presented in more detail at the American Society for Bone & Mineral Research annual meeting in Seattle in October 2015, as well as at the American Society for Gravitational and Space Research (ASGSR) meetings in Alexandria, VA in November, 2015, and NASA's Investigator Workshop (Feb, 2016) with additional data collected by that date.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Abstracts for Journals and Proceedings	Metzger CE, Seidel DV, Allen MR, Hogan HA, Turner ND, Zwart SR, Bloomfield SA. "Modulation of Bone Response to Long-Term Disuse by Iron Overload." Poster Presentation at 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. Conference Proceeding disseminated at the meeting. , Jan-2015
Articles in Peer-reviewed Journals	Avin KG, Bloomfield SA, Gross TS, Warden SJ. "Biomechanical aspects of the muscle-bone interaction." Curr Osteoporos Rep. 2015 Feb;13(1):1-8. http://dx.doi.org/10.1007/s11914-014-0244-x ; PubMed PMID: 25515697; PubMed Central PMCID: PMC4306629 , Feb-2015
Articles in Peer-reviewed Journals	Bloomfield SA, Martinez DA, Boudreaux RD, Mantri AV. "Microgravity stress: bone and connective tissue." Comprehensive Physiology (In press as of August 2015) , Aug-2015